Project 2 Report

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Task 1.

1.a & b $f(x)$ splits completely over $\mathbb{Y}$. We only need to find one root of f in $\mathbb{Y}$ to construct the isomorphism, because we only need one root $f(\omega_i) = 0 \in \mathbb{Y}$ to determine one of the isomorphism $0 = \phi(0) = \phi(f(\omega_i)) = f(\phi(\omega_i)) = f(s(y))$. Since every $\beta \in \mathbb{Y}^\times$ satisfies $\beta^{q-1} = 1$, by Euclid's criterion, $\beta^{(q-1)/2} = \pm 1$. For random a, by shifting our root uniformly in $\mathbb{Y}$, $\beta = (\omega + a)$, then it is a square in $\mathbb{Y}$ when $(\omega + a)^{(q-1)/2} = 1$ with probability $1/2$.

1. randomly sample an element $a \in \mathbb{Y}$.
2. compute $\gcd(f(z), (z + a)^{(p^n-1)/2} - 1)$. Since p is odd, $p^n - 1$ is always even.

n	time (s)
4	0.000
8	0.000
16	0.016
32	0.281
64	6.391

On average we need two attempts to have a success split. The complexity is dominated by the modular exponentiation, which computes $(z + a)^{(p^n-1)/2} \bmod f$ in $Y[z]$ via square-and-multiply. This costs $\mathcal{O}(n \log p)$ modular multiplications, and squaring a degree n polynomial costs $\mathcal{O}(n^2)$ multiplications in $\mathbb{Y}$, each of which cost $\mathcal{O}(n^2)$ in $\mathbb{F}_p$. With $\mathcal{O}(\log n)$ recursive levels the expected cost is $\mathcal{O}(n^5 \log p \log n)$. Other isomorphisms can be obtained by taking the Frobenious conjugates $\{s(y)^{p^i}\}$.

1.c We have $\phi(x) = s(y) := \omega$ which ω^i forms a basis in $\mathbb{Y}$. Writing $y \in \mathbb{Y}$ in this basis reduces to solving a change of basis $[[\omega_0]_y \cdots [\omega_{n-1}]_y] \cdot \mathbf{t} = [y]_y$ since $y = \phi(\psi(y)) = \phi(t(x)) = t(\phi(x)) = t(s(y)) = t(\omega)$. Solving a linear system is $\mathcal{O}(n^3)$.

Task2

Task 2 is almost just a direct implementation following the instruction. My implementation passed the test for $p = 13, n = 64, \alpha = -1$ and $\beta = 1$.

Task3

3.a & 3.b Suppose we are given n tuples $(\mathbf{c}_k, \mathbf{a}_k = \mathbf{m}_k + 2\mathbf{r}_k)$. Let $\omega = \phi(x) = s(y)$. Since ϕ is $\mathbb{F}_p$-linear, $c_k = \phi(a_k(x)) = a_k(\phi(x)) = a_k(\omega)$. Writing all elements in coordinates gives

$$[\mathbf{c}_k]_y = \Phi[\mathbf{a}_k]_x, \qquad \Phi = \begin{bmatrix}[\omega^0]_y & [\omega^1]_y & \cdots & [\omega^{n-1}]_y\end{bmatrix}.$$

Stacking the n samples gives $\mathbf{C} = \mathbf{\Phi}\mathbf{A}$. If $\mathbf{A}$ is invertible, then $\mathbf{\Phi} = \mathbf{C}\mathbf{A}^{-1}$. The polynomial $s(y) = \phi(x) = \omega$ is recovered from the second column of $\mathbf{\Phi}$.

The attack fails when $\mathbf{A}$ is singular. Under the simplifying heuristic that the vectors $\mathbf{a}_k$ behave as uniformly sampled vectors in $\mathbb{F}_p^n$, the probability that $\mathbf{A}$ is non-singular is $\prod_{j=0}^{n-1}(1 - p^{j-n})$.

k	avg time (s)	failures	emperical prob	theoretical prob
1	0.000	20/44	0.454545	0.4398747391
2	0.001	20/103	0.194175	0.2396672624
3	0.001	20/237	0.084388	0.0990771414
4	0.001	20/340	0.058824	0.0622836672
5	0.001	20/634	0.031546	0.0277731527
6	0.001	20/1438	0.013908	0.0151451362
7	0.001	20/1744	0.011468	0.0076923662
8	0.001	20/5276	0.003791	0.0039061907

If we are given only $k < n$ tuples, the equation is underdetermined. To further tackle the attack, we explore the isomorphism other than its linearity. Each tuple can give n polynomial equations in $\mathbb{F}_p$

$$c_k(y) = \phi(a_k(x)) = a_k(\omega) = a_k(s(y)) \pmod{g(y)}.$$

in n unknowns $\mathbf{s} = (s_0, \ldots s_{n-1})$. If the kn equations define a zero-dimensional variety then we can obtain $\mathbf{s}$. We then recover f by constructing the minimal polynomial of $\omega = s(y)$ over $\mathbb{F}_p$.

3.c & d We have much less information in the actual attack. We try to create as many equation as possible with all the known rules for key construction and field constraints.

1. the encoding rule: $c(y) = \phi(a(x)) = (m + 2r)(s(y))$. Although we don't have the smallness premise directly for coefficients of a, we have $a = m + 2r$ structural constraints, with known m and small r_i.

2. a polynomial with small coefficients that satisfies $f(s(y)) = 0$.

3. h_j and r_i are small values sampled from $U_{\alpha,\beta}$. Through Arora-Ge, they satisfy $Q(v) = 0$ where the polynomial is defined as $Q(v) = \prod_{\ell=\alpha}^{\beta}(v-\ell)$.

We inspect the dimension of the resulting variety. The unknowns are (i) the coefficients of s: n, (ii) h: $\lfloor n/2 \rfloor + 1$, (iii) r: n. In total about $5n/2$ unknown variables. The first two construction constraints give $2n$ equations and Arora-Ge gives about $\simeq n/2 + n = 3n/2$ equations. Thus the resulting ideal therefore defines a zero-dimensional variety.

n	# solutions	unique f candidates	time (s)	recovery result
2	1	1	0.000	direct match
3	1	1	0.000	direct match
4	4	2	0.469	direct match

Exhaustive search requires checking $(\beta - \alpha + 1)^{\lfloor n/2 \rfloor + 1}$ candidates for f, which is exponential in n. For the Grobner basis approach, with $5n/2$ unknowns and $5n/2$ equations, and the bound on $d_{\text{reg}} \leq \sum_{i=1}^{m}(d_i - 1) + 1 \simeq \Theta(n^2)$ with $\Theta(n)$ unknowns the complexity is then $\binom{\Theta(n^2)}{\Theta(n^2)} \simeq 2^{\Theta(n \log n)}$ which is super exponential by Stirling's approximation. In general neither of the methods should be preferred.

There could be multiple solutions to this non-linear equation system. The exact solution will appear in the list of solutions. In practice, additional plaintext-ciphertext pairs can verify candidates.

Task4

4.a Because frobenius commutes with isomorphisms.

$$\text{Tr}(\phi(a)) = \sum_{i=0}^{n-1} \phi(a)^{p^i} = \sum_{i=0}^{n-1} \phi(a^{p^i}) = \phi(\sum_{i=0}^{n-1} a^{p^i}) = \phi(\text{Tr}(a)) = \text{Tr}(a)$$

The first equation is directly the definition. The second is given by the commutative frobenius and isomorphism. The third one is given by the linearity of an isomorphism. Since the image of Tr is $\mathbb{F}_p$ and isomorphism fix $\mathbb{F}_p$, we have the last equation.

4.b Consider the dual basis expressed in the original basis $\hat{\omega}_j = \sum_{k=1}^{n} c_{k,j}\omega_k$, then the trace $\langle \omega_i, \hat{\omega}_j \rangle = \langle \omega_i, \sum_{k=1}^{n} c_{k,j}\omega_k \rangle = \sum_{k=1}^{n} c_{k,j}\langle \omega_i, \omega_k \rangle$. Define the trace Gram matrix of the original basis

$$[\mathbf{G}]_{i,j} = \mathrm{Tr}(\omega_i \omega_j).$$

We then solve with the duality condition $\langle \omega_i, \widehat{\omega}_j \rangle = \delta_{ij}$

$$\sum_{k=1}^{n} \mathbf{G}_{i,k} c_{k,j} = \delta_{ij}, \quad \mathbf{G}\mathbf{c}_j = [\mathbf{1}]_j \Rightarrow \widehat{\omega}_j = \sum_{k=1}^{n} (\mathbf{G}^{-1})_{k,j}\omega_k.$$

The complexity is dominated by the trace building $O(n^4)$. There are n^2 entries where each of which cost a $\mathbb{F}_p^n$ product. Matrix inversion costs $O(n^3)$.

4.c&d Given a bunch of $A_k(y)$. We therefore have

$$\langle A_k(y), C_i(y) \rangle = \langle A_k(y), \sum_{j=0}^{n-1} d_j y^j \rangle = \sum_{j=0}^{n-1} d_j \langle A_k(y), y^j \rangle$$

is linear in d_j. We can form an equation with

$$\langle A_k(y), C_i(y) \rangle = \langle \phi(a_k(x)), \phi(c_i(x)) \rangle = \langle a_k(x), c_i(x) \rangle = a_{k,i}$$

which has smallness constraint. Using this constraint, we form the non-linear equation of d_j with Arora-Ge

$$\prod_{\ell=\alpha}^{\beta} \left(\sum_{j=0}^{n-1} d_j \langle A_k(y), y^j \rangle - \ell \right) = 0 \tag{1}$$

Consider in total $k = 1..N$ equations, each of which of degree D_k then citing the handout we use the equation for a semi-regular system, d_{reg} is the index of the first non-positive coefficient. Equation 1 is of degree $D_j = \beta - \alpha + 1$. In total we have N equations. For $n = 32$ and $D = 3$,

$$d_{\mathrm{reg}} = \min \left\{ d : [z^d] \frac{(1-z^D)^N}{(1-z)^n} \leq 0 \right\}$$

N	32	100	1000	10000
d_{reg}	65	11	5	3

To minimize the degree of regularity, we would like enough samples so that the set of equations is already overdetermined. We therefore hope that the number of sampled equations covers the dimension of the vector space

$$N = \binom{n + D - 1}{D}. \tag{2}$$

A sanity check with our example above by plugging in $n = 32$, $D = 3$ we have $N = 5984$ roughly matches the table. For this choice, the degree of regularity is D. The corresponding Gröbner basis complexity is roughly

$$O\left(\left((N\binom{n + D - 1}{D})^{\omega}\right)\right) \simeq O\left(\left(\binom{n + D}{D}^{2\omega}\right)\right).$$

For $D = 3$, this becomes $O(n^{3(2\omega)})$, which is roughly $O(n^{18})$ assuming classical linear algebra operation on matrix elimination factor $\omega = 3$.

We form the set of non-linear equation just as what we derived in the previous subtask. And we form an ideal from the N non-linear equations in variable d_j and then solve by finding its variety. After solving C_i, the solutions we retrieved $\{C_i(y)\}_{i=0}^{n-1}$ contains the dual basis of the basis $(s(y))^i$ of $\mathbb{Y}$. We then solve for the dual of this dual basis using 4.b to retrieve $s(y)$.

The solved dual basis gives us exactly two possibilities up to the sign-flip of the selected $C_1(y)$ we use to recover $s(y)$[1]. With the resulting random $s(y)$ or $-s(y)$, we retrieve back either $f(z)$ or $f(-z)$ through solving the minimal polynomial of $s(y)$ since $\omega = s(y)$ is the root of f in $\mathbb{Y}$. The recovered candidates will match the key construction criterion (monic irreducible, specified shape, coefficient range). Randomly reporting one would give us a fifty-fifty chance of hitting the exact answer.

With the given $\alpha = -1$, $\beta = 1$, the total number of solution we expect would be $2n + 1$, the n basis, flipped sign solutions, and the vector 0. Since $\mathrm{Tr}(A_k C_i) \in -1, 0, 1$, $\mathrm{Tr}(A_k(-C_i)) \in -1, 0, 1$ will also be a possible solution.

This matches the number of solutions found in the experiments:

n	N	# variety sol.	time (s)	recovery result
4	20	9	0.000	matched via sign variant $f(-z)$
8	120	17	0.000	direct match
16	816	33	0.579	matched via sign variant $f(-z)$

Table 1: Experimental results of the search-FFI attack for $p = 2^{16} + 7$, $\alpha = -1$, $\beta = 1$, and $D = \beta - \alpha + 1 = 3$.

Changing to $\alpha = 0$ results in exactly $n + 1$ solutions, the n basis and the zero vector, since $\mathrm{Tr}(A_k C_i) \in 0, 1$ is free from the sign ambiguity. Thus with

the assymmetry coefficient distribution we are guaranteed to obtain the exact secret key. This also reduce largely on the expected N to achieve $d_{\text{reg}} = D = 2$ roughly ten times when plugging in the values back to equation 2.

A remark. To apply the above design to the scheme in Task 2, suppose we have the text-cyphertext pair. The smallness constraint is imposed on $\mathbf{c}_k - \mathbf{m}_k$ and therefore Arora-Ge is

$$\prod_{\ell=\alpha}^{\beta} (\text{Tr}(c_k C_i) - [\mathbf{m}_k]_i - 2\ell) = 0. \tag{3}$$

Choosing $\mathbf{m}_k = \mathbf{0}$ and take div 2 can reduce this back to exactly 4.d. (Suppose we can request the encryption oracle.)

References

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[3] Serge Lang. *Algebra*. Springer Science & Business Media, 2012.